

Contextualizing Knowledge Management Readiness to Support Change Management Strategies

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Abstract

Research on knowledge management (KM) readiness has matured. However, recent organizational structures have emerged which prevent traditional instruments from adequately measuring KM readiness in an organization. For example, service-oriented enterprise (SOE) structure and agile software development techniques are characterized by multiple groups working together with each performing “services” as part of project pattern or process. Each group acts in its own role as a sub-organization with its unique helping patterns and cultures [17]. Traditional instruments which measure aggregate constructs across the entire organization will miss important between-group differences in an SOE that might influence KM readiness. This paper presents a case-based field-study from a large Fortune 500 financial firm transitioning its structure to SOE and considering agile software development methodologies. Survey data was collected along with a series of interviews with key managers and developers. Findings indicate statistically significant differences in KM readiness between groups and the need for alignment.

1. Introduction

Knowledge Management (KM) strategies are maturing including the ability to assess an organization’s readiness for such systems [20, 19, 21]. Critical enablers of KM have been validated and organized in an integrative framework along with organizational processes and performance [14]. These studies and many like them provide frameworks and models for how knowledge can be effectively and efficiently managed within organizations. In particular, knowledge *sharing* (KS) has been identified as a critical process in the creation and application of new knowledge [1]. Because many KM initiatives do not evolve as successfully as their directors had planned, researchers have explored potential barriers and facilitators of KS in organizations [18, 8, 6]. Siemieniuch and Sinclair appropriately described their framework for organizational readiness for knowledge management by saying, *if you would plant roses in the desert, first make sure the ground is wet* [20]. The constructs produced by KM readiness research may help an organization to analyze its preparedness for effective knowledge sharing before a KM system is implemented. This is traditionally accomplished by administering an anonymous survey throughout an

organization which measures critical success factors for KM. However, such techniques may not be appropriate for some emerging organizational structures and processes.

For example, the concept of “agile software development” (ASD) has recently received greater attention from researchers and practitioners alike [15, 12, 10]. The strategy of agile methods is to reduce the cost of change throughout a project [11]. Agility during the lifecycle of the project allows teams to better adapt to shifting demands in user requirements and produce faster rework. But, such dynamics requires the use of expert knowledge during the entire project whereas traditional approaches, like the waterfall method, can still execute using expert knowledge during the project planning phase only [3].

An IT unit planning to make the shift to ASD would benefit from first analyzing their ability to support the kind of expert knowledge sharing necessary to reap the performance benefits of the new process. As stated previously, there is adequate theory to perform such an analysis. However, a key discriminator of ASD is that it is well-matched to small teams and short iterations relative to traditional methods where the entire project is planned and carried out by a single team under a master plan [12].

In Perlow et al.’s [17] analysis used to develop their Nested Theory of Structuration, they discovered three distinct patterns of workgroup interaction within software engineering teams. Each group structure possessed unique interactions between reward structures, helping patterns, and team cultures. This finding allowed them to map the exchange of knowledge and information among and between team members and project leaders. In short, different teams may have different KS norms and practices. Therefore, a blanket survey which measures aggregative KM enabling constructs across all groups will miss important differences in KM readiness *between* teams. Such differences may critically affect the successful implementation of an agile software development process which depends on successful KS between teams.

A Service-Oriented Enterprise (SOE) based on a Service-Oriented Architecture (SOA) is an emerging organizational *structure* which possesses some similar characteristics to the ASD *process*. Brown and Carpenter [4] defined an SOE as *an enterprise that implements and exposes its business processes through an SOA and that provides frameworks for managing its business processes across an SOA landscape*. SOA is characterized generally by its architecture of “services.” A “service” is a standardized component of a business process or project performed by a person, computer, or team. In

the software development process, work gets passed from service to service as it progresses towards completion. Occasionally, rework is required and certain services must be performed again. Throughout this process, knowledge and information must be quickly and efficiently shared between units to remain agile. Similar to ASD, team structures exist which might have differing cultures and patterns of KS. Again, these differences can hinder the agility of an SOE and would go unnoticed by a traditional KM readiness survey.

The paper presents a case study from the business intelligence (BI) unit of a large Fortune 500 financial institution which will be termed ABC Corp for non-disclosure purposes. Based on their need for faster customer response time, ABC Corp is currently exploring agile development methods within an SOE context. Because of the additional knowledge sharing required by these strategies, a survey was administered to measure ABC Corp’s KM readiness. The survey was tailored to detect specific differences between groups.

Results indicate that when KM readiness surveys are used in the context of team-driven structures like SOE and ASD, a group-level analysis is necessary to capture an accurate assessment. Furthermore, such an analysis can better support change management strategies if an organization considers itself “unready” to share and manage knowledge in an agile environment. Practical implications include more targeted and accurate assessments of KM readiness for firms considering similar strategies. This information will improve an organization’s ability to make more actionable decisions based on their KM readiness assessments.

The remainder of the paper is outlined as follows: Section 2 describes the current structure and needs of ABC Corp’s BI unit; Section 3 reviews the agile development methodology and SOA including why these structures might be good fits for ABC Corp’s needs; to assess ABC Corp’s preparedness for these new structures, Section 4 outlines current literature and theory in KM readiness to provide a theoretical base for the survey; also, workgroup helping patterns, which illustrate the flow of knowledge within teams, are discussed to help contextualize the results of the survey; Section 5 presents the survey results; Section 6 discusses limitations of the study and also how results can be contextualized to the appropriate workgroup patterns used to better inform a change management strategy; and Section 7 concludes.

2. ABC Corp

Within ABC Corp is a BI unit which manages the data warehouse. This unit performs typical data warehousing functions including producing reports for large clients based on the data gathered by the warehouse. Most projects require adjustments to database table structures in the data warehouse and some amount of software engineering to either produce the report or give the client access to generate the desired reports from an interface. Projects are typically driven by customer needs and requests. Each project can vary in size from medium to quite large.

ABC Corp had several concerns with the efficiency and effectiveness of their current processes. A questionnaire revealed several general concerns relating to problems in communication, timeliness, teamwork, subject-matter competencies (implicit knowledge), and process organization. The structure of ABC Corp's BI unit consists of several project teams each with a similar set of person resources sufficient to complete a typical project. Recently, ABC Corp switched to a "job shop" or component-based structure where various teams are now responsible for separate components of a single project like data ETL (Export/Transform/Load), development, and testing rather than assigning one team to a complete project lifecycle. Although their organizational structure has changed, ABC Corp continues to follow the waterfall methodology for their software development process.

3. Organizational Structures

With the rapid growth of technology, organizational structures are becoming less of a static concept and more of a "work in progress." Structures are constantly changing to adapt to market demands. During an interview, one ABC Corp team manager noted that customers either do not or can not provide an adequate definition of their own requirements and changes are frequently requested during development. In addition, requirements will frequently change during the development process as a result of new needs or requests from the customer. As a result, project managers end up providing padded cost and time estimates initially to plan for unexpected changes. Traditional project management, which refers to these changes as "scope creep," is more concerned with controlling and preventing change rather than preparing for and expecting it. ABC Corp's difficulties with handling change and new organization structure led to an exploration of ASD methods and an SOA structure.

Project development lifecycles are becoming more characterized by changes in requirements and scope

which doesn't match well to the relatively inflexible plan-driven methods. As Highsmith and Cockburn [11] stated, "the question today is not how to stop change early in a project but how to better handle inevitable changes throughout its life cycle."

3.1. Agile Software Development

The differences between agile and plan-driven methods have been characterized by several authors [15, 12, 10, 3, 11]. Their findings are generally consistent.

Traditional plan-driven methods like the waterfall and spiral models are primarily motivated by the need for predictability and stability. They scale well to large projects and teams and flourish when the environment is stable with infrequent and non-critical change. The more common waterfall method can allow some overlapping of phases, but is typical a linear process. Usually, the project plans, knowledge, and information are well-documented and explicit. A mass of knowledge experts are critical in the early stages of planning and analysis when the project is defined, but needed less as the development and testing are actually carried out according to plan. As a result, client involvement is most important during these early stages. Plan-driven methods do better in a culture with clear policies and procedures. Plan-driven methods assume that projects can be completely defined and will require little, if any, change.

Agile methods like extreme programming (XP), feature-driven development, scrum, and adaptive software development are characterized by the need for change, adaptability, and earlier release. They scale better to smaller team sizes and can handle smaller project sizes than plan-driven methods. Agile methods are characterized by series of short iterations – each like a small waterfall method. Project planning is simpler and knowledge is more tacit. A consistent amount of knowledge experts are required throughout the lifecycle of the project to handle and adapt to changes. Client involvement is required and usually expected throughout the entire project. Agile methods perform well in less-ordered environments with little organizational hierarchy. Agile methods assume small teams using rapid feedback can develop quality software through short iterations of continuous improvement and testing.

3.2. Service-Oriented Architecture and the Service-Oriented Enterprise

Service-oriented architecture has been described as both a technical architecture and also a business modeling concept [4, 9]. Building and integrating SOAs leads to the evolution of an SOE. As a technical architecture, SOA is the standardized infrastructure based on XML and Web Services used to support business processes. Therefore, SOA may also refer to the actual business process supported by a service-oriented infrastructure (SOI). In an SOE, the software development process is basically a choreography of services – like planning, analysis, design, build, test, and implement – which, when performed together, comprise the pattern of a software development project. Each major service may be composed of a series of sub-level services which are performed by a person, computer, or team. The advantages of an SOA stem from the underlying SOI characteristics based on Web Services like speed and agility in both the discovery and invocation of the unit of service. Also, the standardization of the service unit allows for a formal structure for inputs and outputs through each service which can improve workflow speed.

The concept of SOA applies in this research in two distinct ways: 1) ABC Corp's recent change in their BI unit structure from project teams to component teams resembles an SOA where the component teams resemble an architecture of services and 2) an SOE, characterized by its structure of SOAs, can better support an agile development environment consisting of small teams performing services like short iterations in a project development lifecycle.

In summary, agile development methods within an SOA structure carry the theoretical potential to improve speed and adaptability to change (as desired by ABC Corp), but have certain requirements in communication, teamwork, KM, and process structure – all perceived as areas needing improvement at ABC Corp. If they want to change to these organizational structures, ABC Corp needs to first accurately assess their KM readiness.

4. Literature and Theory

As stated previously, theory on KM readiness has progressed in the traditional organizational context. In addition, there is theory to support helping patterns within groups which allows us to make actionable decisions regarding readiness assessments. But, SOA and the trend toward agile development methods present a new structure to examine readiness in. In particular, these new methods and technologies relax the assumption that knowledge sharing and interactions take place *within* groups only. The following subsections outline current literature and

theory in KM readiness and the Nested Theory of Structuration.

4.1. Knowledge Management Readiness

Many studies have attempted to define the constructs which make an organization “ready” to manage knowledge or that cause individuals to intend to share knowledge [2, 20, 5, 19, 21]. Usually, each approach is tested by administering a validated survey instrument. Lee and Choi [14] performed a comprehensive experiment to integrate the many views on KM readiness. Their research examined the relationship between knowledge enablers, processes, and organizational performance in an integrative framework. Knowledge enablers are mechanisms that stimulate knowledge creation, protect knowledge, and facilitate knowledge sharing. Knowledge processes provide the structured coordination to effectively manage knowledge. Organizational performance refers to how well a business is accomplishing its goals in terms of learning, profitability, and other benefits of KM.

Based on established theory and literature, Lee and Choi included organizational culture, structure, people, and IT as KM enablers in their research model. They defined six constructs from a social perspective and one from a technical perspective to measure those enablers:

- 1) *Collaboration* is the degree to which people help each other in their work.
- 2) *Trust* is the faith people have in each other in terms of intention and behaviors.
- 3) *Learning* refers to the degree to which it is encouraged in an organization.
- 4) *Centralization* refers to the locus of decision authority and control within an organization.
- 5) *Formalization* is the point to which decisions and working relationships are controlled by rules, policies, and procedures.
- 6) *T-shaped skills* are the skills in which a person has not only a deep knowledge within a subject (the vertical part of the “T”), but also how that subject interacts across domains (the horizontal part of the “T”).
- 7) *IT support* is simply the degree to which KM can be supported by technology.

Each construct was tested for its affects on the knowledge creation process which consists of socialization (generating new tacit knowledge through social interactions), externalization (converting knowledge from tacit to explicit), combination

(combining key pieces of explicit knowledge into systematic sets), and internalization (embodying explicit knowledge into tacit knowledge) [16].

These constructs provide the basis for a thorough analysis of KM readiness. However, the actual survey items used by the authors were not intended to adequately detect between-group differences in an organization characterized by multiple units interacting with each other like “mini” organizations – which might be found in an SOE or firm practicing agile development methods. In order to make informative decisions regarding team structures and reward systems to improve KM, organizations must adapt proven techniques like Lee and Choi’s instrument to discover how knowledge is currently being sharing and managed between groups.

4.2. Nested Theory of Structuration

Besides the proven KM enablers, there are also patterns of workgroup interaction which affect an organization’s flow of knowledge and information and therefore its readiness for KM. Perlo et al. [17] performed an extensive observational study of three large software engineering firms in different countries. They discovered three distinct patterns of helping within groups which they termed managerial centered, expertise centered, and team centered (Figure 1 – adapted from Perlow et al.). These patterns are based on the roles of project leaders, the roles of software engineers, and the source of knowledge when the need for help arises.

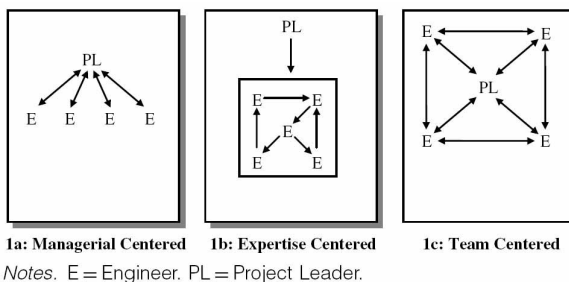


Figure 1 - Patterns of helping within work groups

The managerial pattern (Figure 1a) is characterized by little to no communication among software engineers. In this pattern, the helping interaction is almost completely between engineers and project leaders. Expertise centered patterns are characterized by high helping interaction between engineers. Each team member is known for some kind of expertise that other members rely on for help. The project leader is removed from the day-to-day knowledge-

based technical questions and takes on the role of an overseer rather than provider of information. Finally, the project leader in the team centered pattern is more involved in addressing technical questions. The group is more team-oriented and members can turn to anyone in the group for help.

The mutual reinforcement between human patterns and organizational structures is the premise of Structuration Theory. Perlow et al.’s *Nested Theory of Structuration* posits that organizations are most effective when there is a fit between the institutional context, organizational context, and patterns of workgroup interaction (See Figure 2 – adapted from Perlow et al.). An organizational context might be SOE using agile project development methods for example. In addition, the organizational context takes place within an even larger institutional context. For example, government policies and family values are variables that contribute to organizational context. Each level of nesting helps to shape the structure and patterns of the level without it just as the level within reinforces the structure of the outer level. So, there is a round of structuration within each level.

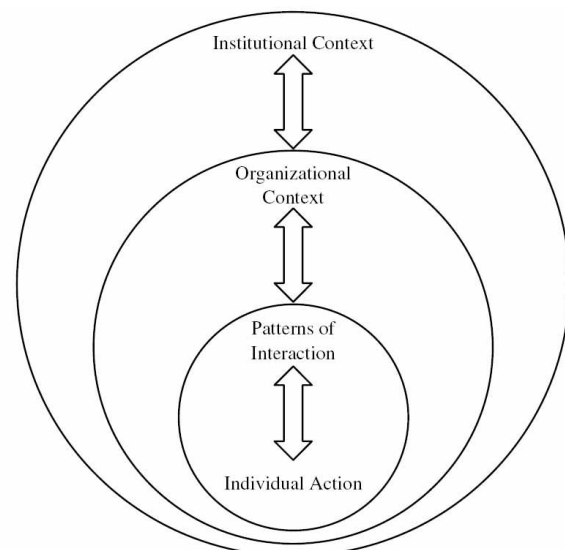


Figure 2 - Nested Theory of Structuration

To understand why it may be inadequate to ask traditional KM readiness questions without taking into account the organizational context and work group patterns, consider a typical survey item. To assess the perceived level of collaboration within an organization, a survey might ask for a user’s level of agreement with the statement, “Our organization members are helpful.” It is possible that more than one type of workgroup pattern exists in a single organization. Assume that data is collected from users in both a managerial centered group and a team

centered group. The user from the team centered group is giving a cumulative response about his or her entire group whereas the user from the managerial group is really only providing feedback on his or her project leader because that is the only person the user solicits help from. If the managerial group project leader is perceived by their subordinates as doing a poor job and is disliked, then responses from that group will be mostly negative, when in reality, maybe only the project leader is unhelpful. Now, consider yourself the director of an IT unit in a large organization trying to assess your KM readiness to switch to agile project development within an SOA. The results from your survey may indicate some groups – the managerial centered groups with poor project leaders – are not optimistic about collaboration. You may decide those groups are not ready for agile development, which requires a high degree of collaboration throughout a project lifecycle, when in reality you may only be suffering from poor project managers.

After assessing KM readiness differences between groups, workgroup interactions should be discovered and used to contextualize the results. Then, these assessments can be used to plan change management strategies considering organizational and institutional contexts.

5. Research Methodology and Results

This research is based on the case study methodology adapted from interpretive research literature [13]. It progressed through a period of 6 months. During this time, the data gathering was partly qualitative through structured and unstructured interviews, but mainly quantitative through the use of a survey instrument. To measure the KM readiness of ABC Corp, we developed a 46-item pilot survey adapted from Lee and Choi to measure their seven KM enablers. There were slight changes in wording to detect within and between group differences. Two of the questions were open-ended to detect additional issues and problems not addressed in the survey.

5.1. Measurement and Data Collection

Surveys were available in paper format and also emailed to 45 employees of ABC Corp's BI unit. This unit is comprised of 5 teams with 7 to 10 members each and a few administrative personnel. To ensure anonymity, we have arbitrarily assigned group numbers and teams will be referred to as group 1, group 2, and so forth. Twenty-seven surveys were

collected yielding a 60 percent response rate. Although each component team is responsible for different functions in the software engineering environment, each team is comprised of similar employee positions. For example each team has a project manager, programmers, and business analysts.

5.2. Results

Figure 3 graphically displays the survey results. The average score on a 7-point Likert scale for each construct is plotted for groups 1 through 5. Each construct was measured by a set of questions adapted from Lee and Choi's validated instrument. Table 1 indicates the actual scores for each group on each KM enabler – collaboration (COL), trust (TRU), learning (LEA), centralization (CEN), T-shaped skills (TKS), formalization (FOR), IT support (ITS) – including the response rate for each group. Survey results indicated several significant differences in KM enablers between groups. Based on a simple t-Tests between groups, group 1 indicated a higher degree of collaboration than either group 4 or 5 ($p = .01$ and $.05$ respectively), group 3 indicated a higher degree of trust than either group 4 or 5 ($p = .02$ and $.05$ respectively), and group 1 rated their centralization as lower than either group 2 or 3 ($p = .001$ and $.01$ respectively).

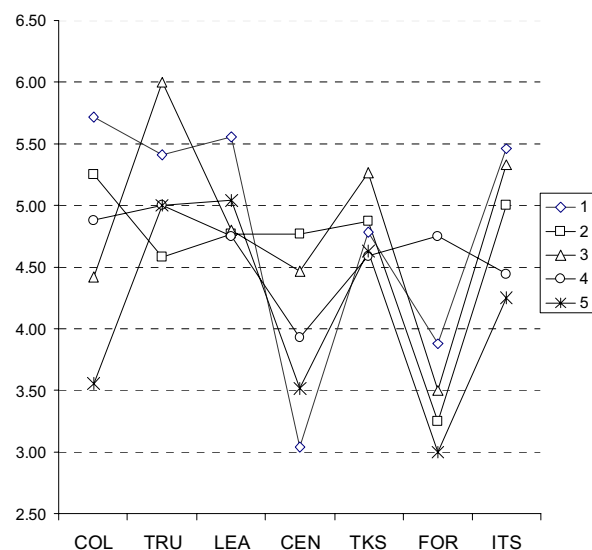


Figure 3 - Results by group on KM enablers

Table 1 – Group means and response rates

Group	COL	TRU	LEA	CEN	TKS	FOR	ITS	Responses	Rate
1	5.72*	5.41	5.56	3.04*	4.78	3.88	5.46	8	73%
2	5.25	4.58	4.77	4.77	4.87	3.25	5.00	3	43%
3	4.42	6.00*	4.80	4.47	5.27	3.50	5.33	3	38%
4	4.88	5.00	4.75	3.93	4.59	4.75	4.44	4	50%
5	3.56	5.00	5.04	3.52	4.63	3.00	4.25	5	56%
Other	3.33	4.90	5.20	4.00	3.92	3.71	3.13	4	N/A
Overall	4.69	5.17	5.12	3.71	4.66	3.63	4.91	27	60%

Scores are based on a Likert scale, 1 = lowest and 7 = highest

* Indicates a p-value <= .05

These results are consistent with data collected prior to the survey. In an interview near the beginning of the project, the managing director and cost manager of the BI unit (both have more than 5 years experience in their positions at ABC Corp) revealed that group 1 had consistently been the top performer both before and after the BI unit's structural change to component teams. Their subjective judgment agrees with the survey results which indicate that group 1 scores generally higher than other groups on KM enablers which Lee and Choi determined to influence the knowledge creation process which in turn is related to performance.

6. Discussion and Limitations

The purpose of the survey was to capture differences in KM readiness between groups if differences indeed exist. Based on the survey results and reports from management, the instrument accomplished its purpose. In addition, besides validating the perceptions from management, the survey identified the specific KM enablers in which groups differ. This supports more targeted change strategies to improve readiness for KM. However, this study has several limitations. First, the case study approach, while still a valid method of interpretive research, has limited generalizability because it only represents the results of one firm [13]. Second, even for a case study, the total number of responses was limited and may not represent an accurate assessment for each group – particularly in the case of groups 2 and 3 from which only three surveys were collected. Finally, the study places KM readiness in the context of assessing an organization transitioning its structure to SOE and implementing agile development methods. While we propose KM readiness should be addressed at the group level in ABC Corp's context, this may not be appropriate for all firms considering an SOE structure. As stated previously, the services performed by an SOA may be provided by an individual or web service and not necessarily by a team. This is also why the specific organizational

context should be considered when assessing KM readiness.

6.1. Contextualizing Survey Results

When specific between-group differences are discovered, they should be considered in the context of the particular helping pattern the group(s) of interest follow. For example, survey results indicate that group 1 is significantly less centralized than groups 2 and 3. This information may help to confirm a managerial centered workgroup structure in group 2 and 3 and a team or expertise centered structure in group 1. If ABC Corp is planning to use agile development methods, which require more empowerment in team members to handle changes, they might try to alter the workgroup pattern of groups 2 and 3 as part of their change management strategy. Without examining the KM readiness at the group level, the organization would appear to have an average degree of centralization with a score of 3.71 out of 7. If the company then implemented an agile development strategy, they might find later that groups 2 and 3 do not perform as well in an environment that favors decentralized decision making and individual empowerment.

Another example stems from the difference in collaboration between group 1 and 5. The overall score for perceived collaboration is 4.69. Without examining the differences between groups, a company may determine that their degree of collaboration is sufficient and change their structure to an SOE. During a software development project within that SOE, work progresses as it passes from service to service. Occasionally, the workflow may follow multiple paths at once. If group 5 is performing a service in one path and discovers a bug or some other critical need for rework and is less willing to collaborate and share knowledge with groups performing services in other paths of the workflow, the agility of the process could be hindered.

In summary, we have used the example of ABC Corp considering a change to SOE and agile development to argue that KM readiness assessment should be adapted to perform a group-level analysis. We have also explained how the survey results of ABC Corp might be placed in the context of the workgroup helping patterns to better understand the results. The next step is to use these contextualized results to improve change management strategies.

6.1. Change Management Strategy

Demirors and Frailey [7] defined a “horizontal change approach” to software process improvement. Their horizontal approach was motivated by the lacking ability of vertical approaches to quickly bring about desired change strategies. Vertical approaches focus on how to model, control and enact certain processes or improvements in a top-down fashion. They are similar to the waterfall life-cycle model where knowledge is at the top and less-skilled workers at the bottom. In the vertical approach, the bottom workers generate data from their activities which is analyzed by top management and returned to them in the form “task prescriptions.” However, software development is a knowledge based environment. The vertical approach does not work well for software process improvement where the knowledge is generated and in the greatest amounts at the bottom and is more difficult to analyze at levels other than where it was created.

The horizontal approach emphasizes an environment that utilizes the expertise of all knowledge workers to achieve the change strategies. It is better structured to successfully accomplish change strategies in software development processes – especially as an organization moves to agile development methods where knowledge is shared and managed horizontally. Of the six principles of horizontal change approaches identified by Demirors and Frailey, three can be particularly supported by our findings:

1. Processes should enable the surfacing of assumptions and expectations in an organization: gaps between the inherent assumptions and expectations of individuals cause major process inefficiencies.

The main goal of this research is to increase the surfacing of assumptions made by individuals by measuring their perceptions between groups in a survey and placing them in their proper workgroup and organizational context. This information can be used to reduce the gaps (by aligning differences) and better enable change.

2. Diverse communication should occur among individuals and spontaneous distribution of knowledge should occur irrespective of hierarchy.

A KM readiness assessment between groups such as the pilot survey performed in ABC Corp can discover where communication is being frustrated and establish controls to improve it. For example, meetings can be coordinated where necessary communication is not taking place. The type of

helping patterns of the groups involved would affect who the most appropriate communicators are between groups.

3. “More control occurs from less control”: knowledge workers should be empowered to take responsibility for measuring their processes, identifying their problems, improving their processes and providing feedback for the processes they consume.

This principle suggests an expertise or team centered workgroup pattern may facilitate change better in a software development group because they are characterized by more independence from the project leader. Understanding those patterns and being able to detect them from a more targeted survey may help improve process change strategies.

7. Conclusion

From the case analysis of ABC Corp, this study demonstrates that traditional KM readiness assessments should be adapted to detect group-level differences if they are to be effective for emerging team-driven structures like SOE and ASD methods. In addition, these assessments need to be considered in the context of the helping patterns that each group follows. Doing so can help an organization create a better “fit” between the organizational context and workgroup patterns of interaction which will, in turn, improve the effectiveness of the organizational structure. In addition, the contextualized KM readiness assessments can improve change strategies for software process improvement.

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